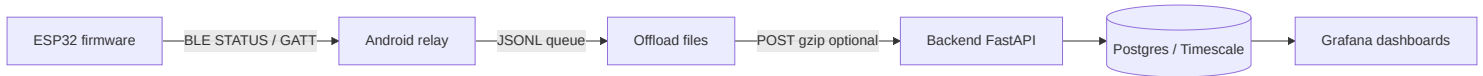


Metrics pipeline — what we report today

Schema envelope for all cloud ingest: `imu.ingest.v1`.

Base URL (production): `https://apps.f0xx.org/app/good_vibes/v1/...`

1. End-to-end path



Heavy math (FFT, band RMS, correlation) runs **on-device** for vibro verdicts; phone FFT is used for optional spectrum upload. Pending rows are queued on the phone when cloud is off or BLE-only.

2. Metrics successfully reported to the backend

2.1 Verdicts — `POST /v1/ingest/verdicts`

Field	Source	ESP STATUS key	Stored
-----	-----	-----	-----
<code>seq , ts_ms</code>	Phone at export	<code>s</code>	<code>verdicts.seq , ts_ms</code>
<code>level</code>	Vibro verdict	<code>vd</code> (0=OK, 1=WARN, 2=ALERT)	<code>verdicts.level</code>
<code>rms , peak</code>	Live capture	<code>vrms , vpeak</code>	columns
<code>corr , rms_delta</code>	vs reference	<code>vcorr , vrmsd</code>	columns
<code>pct , voltage</code>	Battery	<code>pct , v</code>	columns
<code>power_profile</code>	Power manager	<code>pp</code>	column
<code>chip_temp_c</code>	IMU / SoC	<code>tc</code>	column
<code>band_corr , band_delta_max</code>	64-pt FFT bands	<code>bcorr , bdmx</code>	<code>raw_json</code> + API expand

bands[4]	Band RMS	bnd[] or compact b16[]	raw_json
edge_crest, edge_zcr_hz, edge_hf_ratio	Time-domain	cr, zcr, hfr	raw_json
edge_score, edge_risk	Server computed	—	raw_json on ingest
session_seq	Offload spool	psess	raw_json
cap_mix_sec	Capture schedule	capmix	raw_json

Dedup: (device_id, seq).

2.2 Spectra — POST /v1/ingest/spectra

Field	Source	Notes
-----	-----	-----
seq, sample_hz, bin_hz	Phone VibroFft on RAW IMU	User-triggered FFT menu
bins[] (128)	Magnitude spectrum	DC bin dropped
peak_hz, peak_mag	Peak search	
axis	"mag"	

2.3 Crashes — POST /v1/ingest/crashes

Field	Source
-----	-----
reason, pc, exccause, excvaddr	Crash ring GATT
thread_name, fw_version, reset_reason, uptime_ms	
backtrace[], detail	Symbolicated server-side when ELF staged

2.4 Battery bench — POST /v1/ingest/battery_bench

Field	Source
-----	-----

session_id, seq, ts_ms	STATUS bsid, bseq + phone clock
voltage, pct, trend_v, src	STATUS v, pct, tr, p
cpu_mhz, imu_hz, render_hz, chip_temp_c, uptime_ms	STATUS / bench snapshot
session_started_ms, session_stopped, label	Phone session metadata
est_ma	Server computed from dV/dt (500 mAh default)

2.5 Clock events — POST /v1/ingest/clock *(v124+)*

Field	Source
-----	-----
drift_ms, corr_ms, tz_min, unix_sec, src	STATUS clkd, clkc, tz, clku, clksrc

2.6 Reference profiles — PUT /devices/{id}/reference_profiles/{slot} + POST /v1/ingest/reference_profiles *(v124+)*

Field	Source
-----	-----
slot, name, duration_ms, sample_hz	Device reflist GATT
bands[4]	Ref profile band RMS
raw.mag_f16 / raw.bins	Compact magnitude fingerprint (optional)
active	Active slot flag

2.7 Device config — POST /v1/ingest/config

188-byte device_config_v1 blob + JSON mirror; not part of uploadAll() .

2.8 Query-only / metadata

API	Purpose
-----	-----
GET /devices/{id}/verdicts	History + edge fields

GET /devices/{id}/trend	Rolling trend
GET /devices/{id}/battery_bench/sessions	Bench sessions
machines , sensors	Registry (profiles.txt vision)

3. Not persisted (by design today)

Data	Why
-----	-----
Live IMU RAW/COMPUTED/SCENE batches	BLE NOTIFY → UI only
WiFi net STATUS JSON	Local wizard
Full reference mag[512] waveform	Flash-only; cloud gets bands + optional f16 digest

4. Compression & compact formats *(v124+)*

4.1 Design rule

Semantic layer unchanged (verdict levels, bands, spectra). **Encoding layer** may compact:

Layer	Format
-----	-----
BLE STATUS bands	b16 : 4× IEEE754 float16 as JSON uint16 array (8 bytes vs ~40 bytes text bnd)
Phone JSONL	MetricsCompact : optional gzip blob file verdicts.jsonl.gz
HTTP ingest	Content-Encoding: gzip accepted on POST bodies
Flash verdict store	4 band RMS already float32; mag refs use f16 in cloud upload payload

4.2 Float16 band encoding

```
f16 = round(clamp(band_rms, 0, 65504) to float16)
on wire: 4 × uint16 little-endian
```

Android and firmware share metrics_f16 helpers.

5. DSP usage *(v124+)*

Location	Engine
-----	-----
Firmware band RMS / ref FFT	<code>vibro_fft.c</code> — radix-2 Hanning RFFT (esp-dsp hook via <code>CONFIG_VIBRO_ESP_DSP</code>)
Android spectrum	<code>VibroFft.kt</code> — same algorithm; esp-dsp N/A on phone
Time-domain edges	<code>vibro_features.c</code> — no FFT

All vibro **verdict** paths call `vibro_band_rms_compute_series()` → shared FFT core.

6. Reference calibration wizard *(Android v1.17+)*

- 1. Confirm **reset all reference slots** (gas turbine recalibration after vibro tube install).
- 2. For each reference **N of M** ($M \leq 5$, user-chosen): record 5–15 s on-device.
- 3. On completion: upload profiles to backend; device listens for shakes.
- 4. Anomaly (verdict WARN/ALERT) → **red** WS2812 on GPIO38; **5 consecutive OK series** → green.

Appendix A — Extension plans & status

Item	Status	Notes
-----	-----	-----
Verdict edge fields in cloud	Done	<code>raw_json</code> + <code>_verdict_out()</code>
<code>session_seq</code> , <code>cap_mix_sec</code> upload	Done v124	CloudUploader forwards
Clock ingest table	Done v124	<code>clock_events</code>
Reference profile device → cloud	Done v124	Wizard + POST ingest
GATT compact <code>b16</code> bands	Done v124	STATUS when bands valid
JSONL gzip pending queue	Done v124	<code>MetricsCompact.kt</code>
HTTP gzip ingest	Done v124	FastAPI middleware
esp-dsp linked FFT	Hook only	Enable <code>CONFIG_VIBRO_ESP_DSP</code> when module in west

POST /ingest/batches IMU waveforms	Planned	ROADMAP
Mesh / Coded PHY metrics path	Planned	See BLE research notes
Sticker asset beacon	Planned	Separate product mode
Net/WiFi status persist	Low priority	Ops debug
Server-side reference → device download	Planned	Provision ideal profiles

Appendix B — Key files

Layer	Path
-----	-----
Protocol	zephyr/app/common/ble_imu_protocol.h , ImuProtocol.kt
Vibro DSP	zephyr/app/common/vibro_fft.c , vibro_band_rms.c
LED anomaly	zephyr/app/handshake/src/vibro_led.c
Android upload	CloudUploader.kt , OffloadExporter.kt , MetricsCompact.kt
Ref wizard	VibroRefWizardActivity.kt
Backend	backend/app/api.py , models.py , schemas.py